

MAATRICARE AI: A MULTILINGUAL MATERNAL HEALTH RISK MONITORING SYSTEM

An AI-Powered Clinical Decision Support System for Rural Care

Presented By: Group 6

Under Supervision of
Prof. Esha Choudhuri

GROUP MEMBERS DETAILS

Name	Enrollment Number	Department
Rohan Hazra	12024002028203	CSE (AI&ML)
Prapti Acharya	12024002028037	CSE (AI&ML)
Souryadeep Dutta	12024002028220	CSE (AI&ML)
Fardeen Khan	12024002028024	CSE (AI&ML)
Swapnaneel Dey	12024002028229	CSE (AI&ML)

TABLE OF CONTENTS

Section	Slide Number
CHAPTER - 1:Problem Statement	4
CHAPTER - 2:Proposed Solution	5
CHAPTER - 3:Literature Survey	6
CHAPTER - 4:Technology Stack	7
CHAPTER - 5:Methodology & System Architecture	8
CHAPTER - 6:Database Architecture	9
CHAPTER - 7: ML Risk Pipeline: Data to Decision	10
CHAPTER - 8: Deployment & Expected Impact	11
CHAPTER - 9:Conclusion & Future Work	12

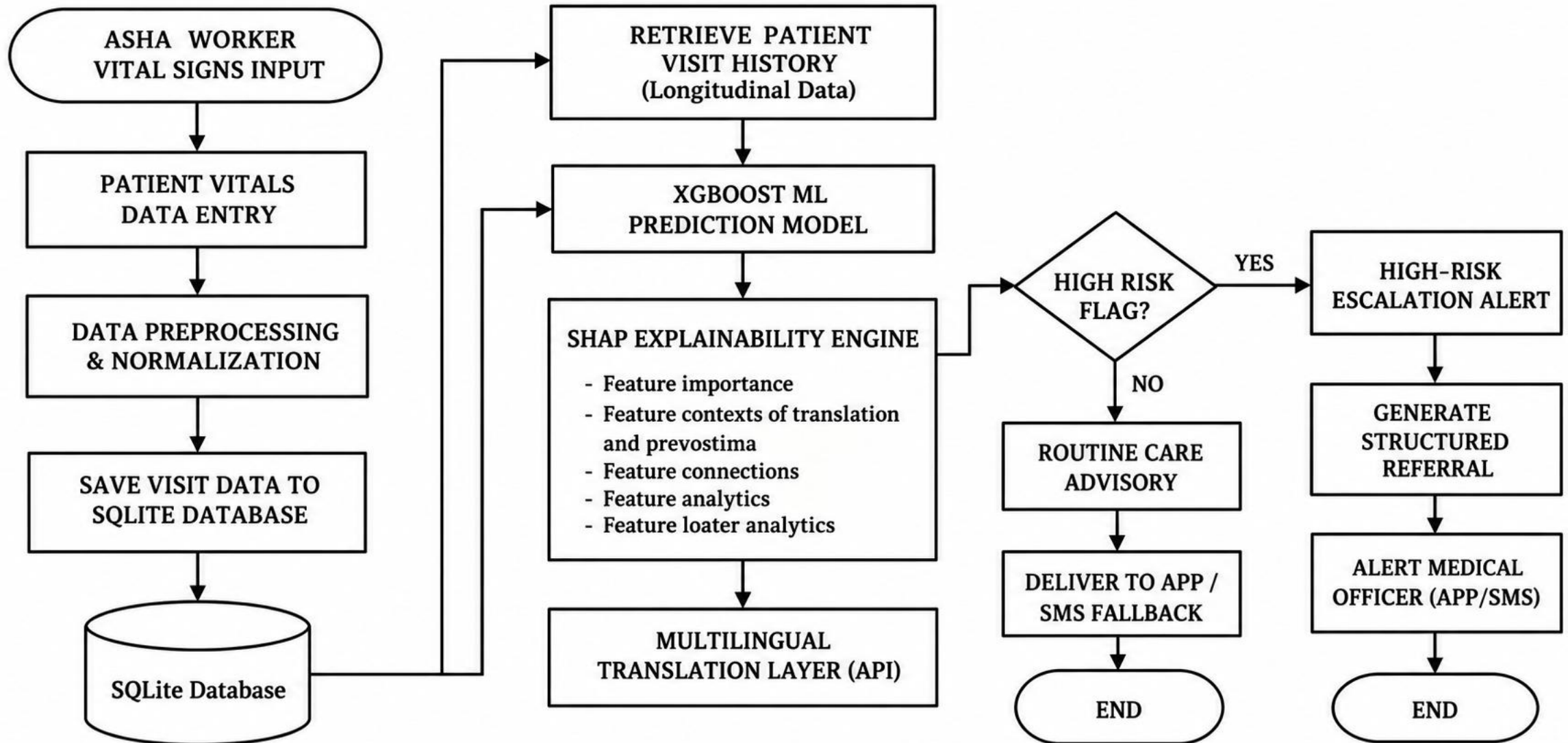
- Maternal complications are frequently identified only after they escalate, especially in rural and resource-limited areas
- Health information systems rarely communicate in the patient's or frontline ASHA worker's local language, creating significant communication gaps
- Existing tools score risk on a one-time snapshot basis, missing crucial data on how a patient's condition trends across multiple visits .
- Rural regions often lack reliable internet, making "always-online" healthcare systems impractical for frontline use
- Nearly 287,000 preventable global maternal deaths occur annually, emphasizing the critical need for timely detection tools



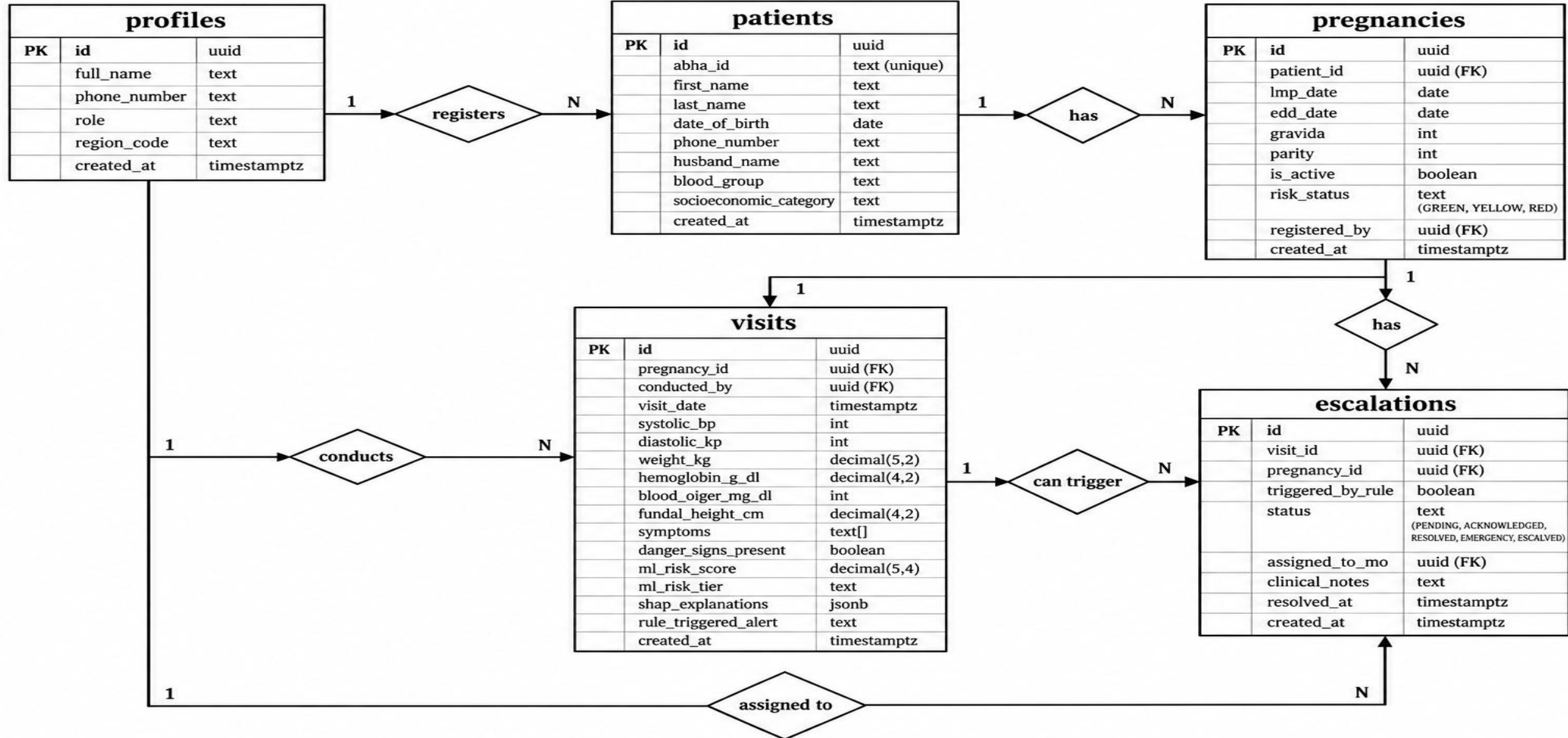
- **MaatriCare AI** is a multilingual maternal health risk monitoring system that converts routine visit data into clear, actionable, local-language risk guidance for frontline workers
- The solution utilizes **Explainable Risk ML** (XGBoost + SHAP) to classify risk (Low / Medium / High) and provide transparent reasoning, building trust
- It features an integrated **Multilingual Output** layer that delivers prediction results in the worker's local language (Bengali, Hindi, English)
- The system uses **Trend-Aware Monitoring** to track visit history across trimesters and trimesters, rather than just scoring a single snapshot
- It is designed to be **Offline / SMS-Ready**, ensuring that data entry, risk scoring, and critical high-risk alerts function fully in areas with poor internet connectivity

Study (Author & Year)	Methodology / Approach	Key Result	Gap Identified
Sarker et al., 2025 (Healthcare Analytics)	PCA-enhanced XGBoost + SMOTE-ENN	97.7% accuracy on maternal dataset	No multilingual or offline deployment focus for rural areas
Altaie & Ibrahim, 2026 (DHJ)	Leakage-regulated hybrid XGBoost	ROC-AUC 0.911 with calibration	Lacks explainability (SHAP) layer for frontline workers
Predicting Maternal Risk, 2024 (BMC Preg. & Childbirth)	Comparative models (LR, RF, KNN, XGB)	Model ranks shift post-PCA	Single-point scoring only; no visit trending
Risk via Boosting, 2025 (Springer)	UMAP dimensionality + XGBoost	Improved class separability	Ignores language and accessibility limitations

Layer	Technology	Purpose
Frontend	HTML	Structure of the web interface
	CSS	Styling and responsive UI
	JavaScript	User interaction and frontend logic
Backend	Python	Core backend and AI processing
	Flask	API/backend server
	Pandas	Data processing and manipulation
	XGBoost	Maternal risk prediction
	Joblib	Saving/loading the trained ML model
Explainable AI	SHAP	Explaining why a patient is classified as high/low risk
Database	SQLite	Storing patients, pregnancies, visits and escalation records
Translation	Google Translation API	Multilingual recommendations
Communication	SMS	Alert/fallback communication for low-connectivity areas



DATABASE ARCHITECTURE



1. Input — Routine Visit Data

- Captures vitals at each visit: Systolic/Diastolic BP, Weight, Hemoglobin, Blood Sugar, Fundal Height.
- Includes reported symptoms (e.g., headache, swelling, blurred vision) and flags danger signs.

2. Preprocessing

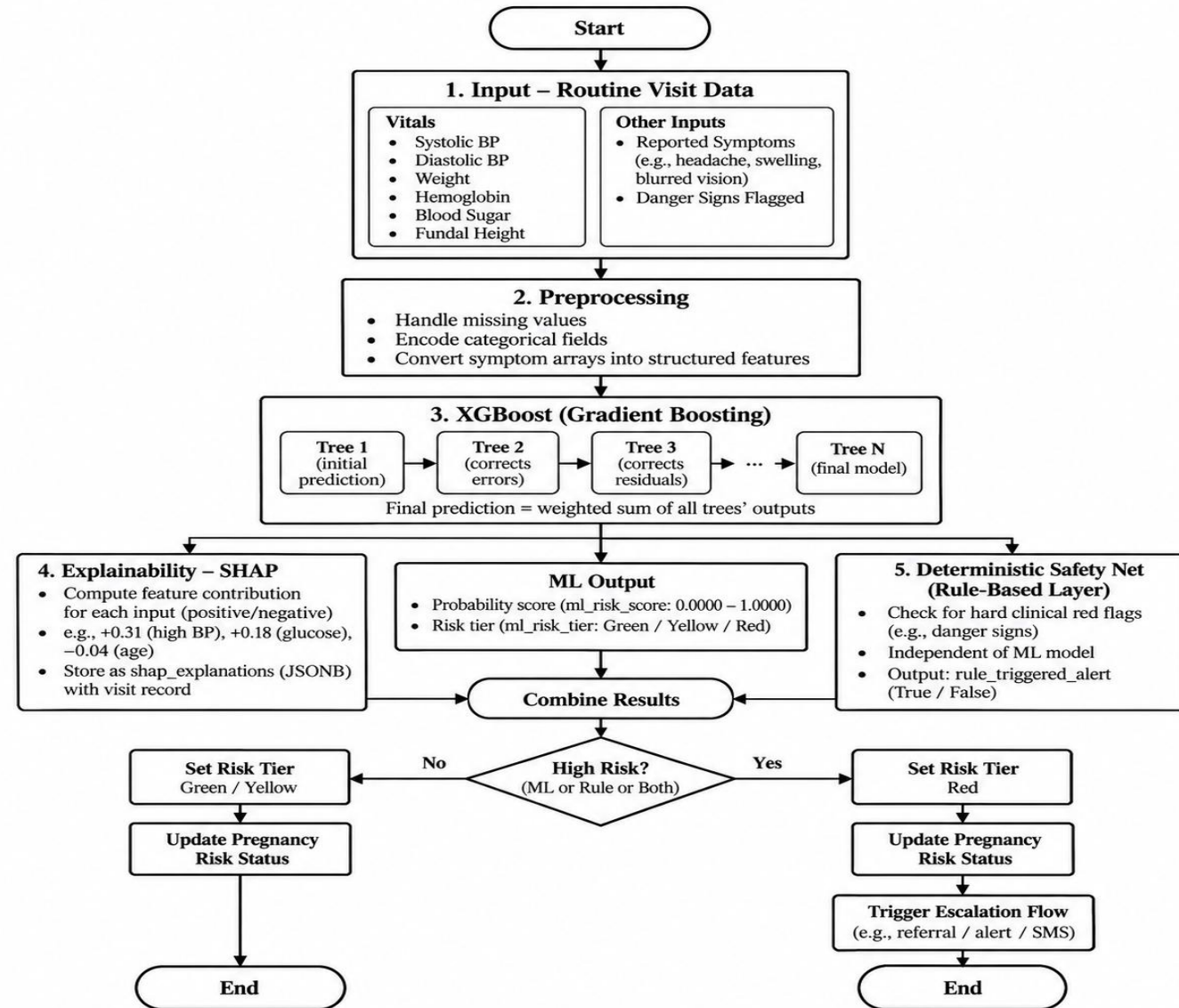
- Handles missing values and encodes categorical fields.
- Converts symptom arrays into structured features for model ingestion.

3. Core Model — XGBoost (Gradient Boosting)

- Builds an sequential ensemble of decision trees; each tree corrects the errors (residuals) of previous ones.
- Final prediction is a weighted sum of all trees' outputs.
- Output: Probability score (0.0000–1.0000) mapped to a Green / Yellow / Red tier.

4. Explainability — SHAP

- Assigns a contribution value (positive or negative) to each input feature for every prediction.
- Shows exactly "why" the model flagged a patient (e.g., "+0.31 from high BP, -0.04 from age").
- Stored as JSONB format alongside the visit record.



DEPLOYMENT & EXPECTED IMPACT

Deployment	Expected Outcome
Pilot deployment with district health authorities and PHCs	Real-world validation of MaatriCare AI in rural maternal healthcare settings
Train ASHA and ANM workers	Improved adoption of AI-assisted maternal risk monitoring and decision support
Offline-first app with SMS escalation	Continued access and timely alerts even in low-connectivity areas
Integration with ABDM and facility EMRs	Better continuity of patient records and easier coordination between healthcare facilities
State-level rollout across high-burden districts	Wider coverage, earlier identification of high-risk pregnancies, and improved maternal healthcare outcomes

Conclusion

MaatriCare AI successfully combines Explainable AI (SHAP), Multilingual Communication, and Continuous Risk Monitoring into a cohesive system. It acts as a critical decision-support tool, empowering ASHA workers to intercept complications early, specifically designed for the realities of rural offline environments.

Future Work

- **Production Deployment:** Transitioning backend to FastAPI and containerizing with Docker for robust PHC scaling.
- **Federated Learning:** Training the model across diverse geographical regions without centralizing sensitive patient records.
- **National Scale:** Direct integration with national maternal health registries and the ABDM framework.
- **Language Expansion:** Broadening the translation API to support 30+ regional Indian languages with voice-input capabilities.